# EXP:03 MATLAB-Based Modelling of VSB Modulator and Demodulator for Analog Television Communication Applications

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clc;
clear;
close all;

%% VSB-SC MODULATOR AND DEMODULATOR

%% 1. Parameters
Fs = 500000;
T = 0.02;
t = 0:1/Fs:T-1/Fs;
N = length(t);

fc = 100000;      % Carrier frequency
Ac = 1;           % Carrier amplitude

fm1 = 2000;
fm2 = 5000;
fm3 = 8000;

%% 2. Message / TV Video Signal
m = 0.5*cos(2*pi*fm1*t) + ...
   0.3*cos(2*pi*fm2*t) + ...
   0.2*cos(2*pi*fm3*t);

m = m / max(abs(m));

%% 3. Carrier
carrier = Ac*cos(2*pi*fc*t);

%% 4. DSB-SC Modulation
dsb_sc = m .* carrier;

%% 5. Frequency Axis
f = (-N/2:N/2-1)*(Fs/N);

%% 6. DSB-SC Spectrum
DSB = fftshift(fft(dsb_sc));

%% 7. VSB Filter
Bm = fm3;
Bv = 2000;

H = zeros(size(f));

% Positive frequencies
H(f >= fc & f <= fc + Bm) = 1;
H(f >= fc-Bv & f < fc) = 0.5;

% Negative frequencies
H(f <= -fc & f >= -fc-Bm) = 1;
H(f < -fc & f >= -fc+Bv) = 0.5;

%% 8. VSB Signal
VSB_spectrum = DSB .* H;

vsb = real(ifft(ifftshift(VSB_spectrum)));

%% 9. Add Noise
SNR_dB = 30;

signal_power = mean(vsb.^2);
noise_power = signal_power / (10^(SNR_dB/10));

noise = sqrt(noise_power) * randn(size(vsb));

received_vsb = vsb + noise;

%% 10. Coherent Detection
local_carrier = 2*cos(2*pi*fc*t);

detector_output = received_vsb .* local_carrier;

%% 11. Low Pass Filter
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cutoff = Bm;

[b,a] = butter(6, cutoff/(Fs/2));

recovered_signal = filtfilt(b,a,detector_output);

%% 12. Normalize
recovered_signal = recovered_signal / max(abs(recovered_signal));

%% 13. Performance Analysis

MSE = mean((m - recovered_signal).^2);

R = corrcoef(m,recovered_signal);
Correlation = R(1,2);

signal_power = mean(m.^2);
error_power = mean((m-recovered_signal).^2);

SNR = 10*log10(signal_power/error_power);

%% 14. Display Results

fprintf('\n====================================\n');
fprintf('       VSB SYSTEM RESULTS\n');
fprintf('====================================\n');

fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);
fprintf('Message Bandwidth       = %.2f kHz\n',Bm/1000);
fprintf('Vestigial Bandwidth     = %.2f kHz\n',Bv/1000);
fprintf('VSB Bandwidth           = %.2f kHz\n',(Bm+Bv)/1000);

fprintf('\nPerformance:\n');
fprintf('MSE                     = %.6f\n',MSE);
fprintf('Correlation Coefficient = %.4f\n',Correlation);
fprintf('SNR                     = %.2f dB\n',SNR);

fprintf('====================================\n');

%% 15. Plot Original Signal

figure;
plot(t*1000,m,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Original TV Video Signal');
grid on;
xlim([0 2]);

%% 16. Plot DSB-SC Signal

figure;
plot(t*1000,dsb_sc);
xlabel('Time (ms)');
ylabel('Amplitude');
title('DSB-SC Signal');
grid on;
xlim([0 0.5]);

%% 17. DSB-SC Spectrum

figure;
plot(f/1000,abs(DSB)/N);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('DSB-SC Spectrum');
grid on;
xlim([-120 120]);

%% 18. VSB Filter Response

figure;
plot(f/1000,H,'LineWidth',1.5);
xlabel('Frequency (kHz)');
ylabel('Gain');
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```
title('VSB Filter Response');
grid on;
xlim([-120 120]);

%% 19. VSB Spectrum

figure;
plot(f/1000,abs(VSB_spectrum)/N);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('VSB-SC Spectrum');
grid on;
xlim([-120 120]);

%% 20. Received VSB Signal

figure;
plot(t*1000,received_vsb);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Received VSB Signal');
grid on;
xlim([0 0.5]);

%% 21. Product Detector Output

figure;
plot(t*1000,detector_output);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Product Detector Output');
grid on;
xlim([0 2]);

%% 22. Recovered Signal

figure;
plot(t*1000,recovered_signal,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Recovered TV Signal');
grid on;
xlim([0 2]);

%% 23. Original vs Recovered

figure;

plot(t*1000,m,'LineWidth',1.2);
hold on;

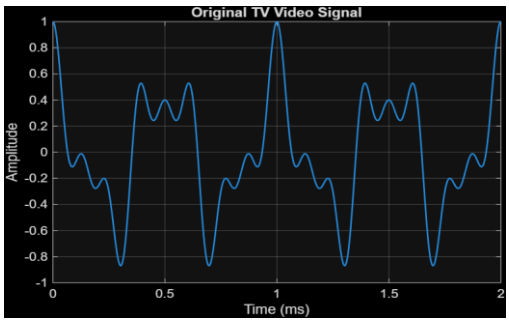
plot(t*1000,recovered_signal,'--','LineWidth',1.2);

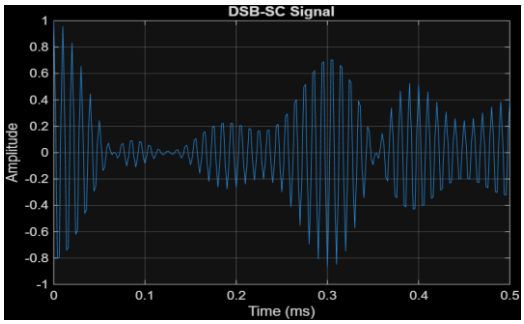
xlabel('Time (ms)');
ylabel('Amplitude');

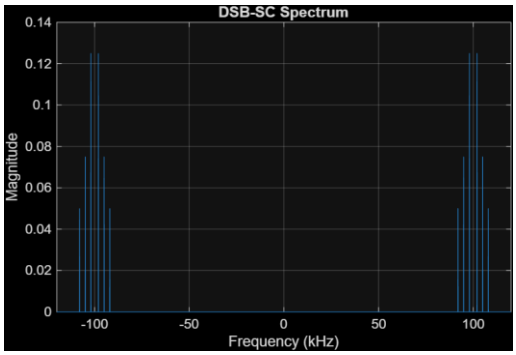
title('Original vs Recovered TV Signal');

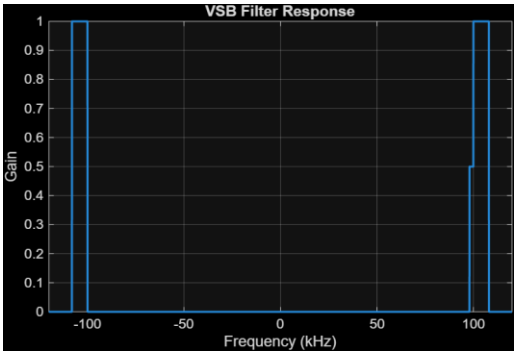
legend('Original Signal','Recovered Signal');

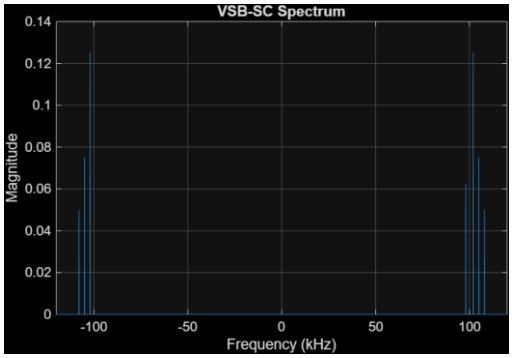
grid on;
xlim([0 2]);
```

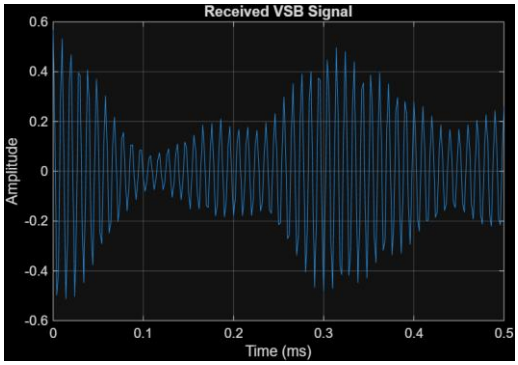


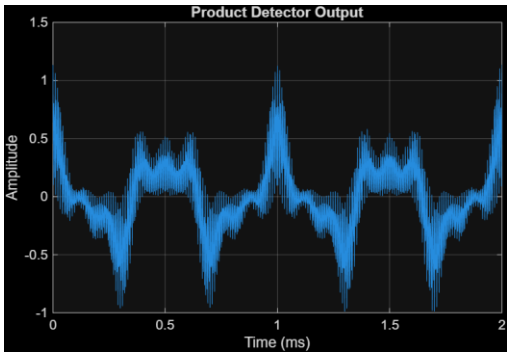



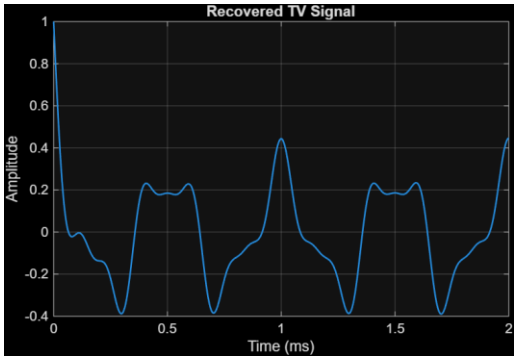

DSB-SC Spectrum
Magnitude
Frequency (kHz)

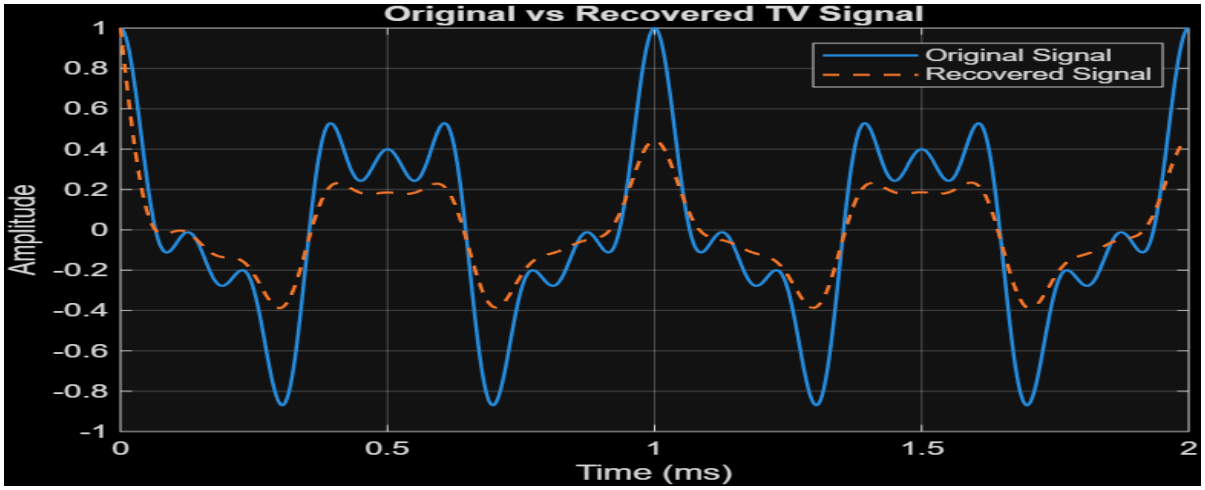

VSB Filter Response
Gain
Frequency (kHz)


VSB-SC Spectrum
Magnitude
Frequency (kHz)


Received VSB Signal
Amplitude
Time (ms)


Product Detector Output
Amplitude
Time (ms)


Recovered TV Signal
Amplitude
Time (ms)


Original vs Recovered TV Signal
Original Signal
Recovered Signal
Amplitude
Time (ms)